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Predicting the Distribution of Petrophysical Properties in Carbonate Reservoirs Using Process-Based Forward Modeling

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Abstract

In carbonate reservoir studies, the results of traditional geostatistical distribution of petrophysical properties in reservoir models remain highly uncertain. Process-based forward modelling is a promising way of taking into account all the geological processes forming and then affecting the pore network, and thus of predicting the distribution of petrophysical properties. Here is proposed a global and efficient forward modeling workflow that jointly simulates the genesis of both geological and petrophysical properties of carbonate reservoirs, from deposition to the present day. Computational efficiency is ensured by focusing on the processes that mostly affect the heterogeneity of petrophysical properties. Given the high impact of early diagenesis, depositional processes and synsedimentary diagenetic processes are simulated jointly rather than sequentially. Finally, later diagenesis processes, such as karst formation, which require consideration of fluid flows in the reservoir, itself deformed and fractured during its structural history, are simulated.

Initial simulations have yielded realistic and promising results. Simulating the various synsedimentary processes makes it possible to produce models whose overall organization of petrophysical properties is linked not only to sedimentary facies, but also to changes in environmental conditions, such as hydrological and diagenetic zoning. As well, the simulation of dissolution processes linked to correctly represented paleofluid circulations makes it possible to produce realistic models with heterogeneous triple-medium characteristics.

Because it preserves geological causality and not simply interpolates between conditioning data, the proposed software enables the present days petrophysical properties of carbonate reservoirs to be actually predicted. It therefore disrupts conventional workflows and allows the data to be used to falsify modeling assumptions, reduce significant initial uncertainty and calibrate predictions.

Introduction

Reservoir modeling is primarily aimed at predicting the petrophysical properties of reservoirs, to perform fluid dynamics numerical simulation. This challenging task typically begins by analyzing and clustering geological properties like facies or rock-types. These properties, and subsequently the corresponding

reservoir properties such as porosity and permeability, are then distributed in a 3D model according to geostatistical rules conditioned by well data. Geological modelling thus serves as an essential intermediate step to help grasping the heterogeneous distribution of underground properties, especially petrophysical properties.

However, this conventional step-by-step workflow generally yields highly uncertain results. This uncertainty is most often attributed to two main factors: the scarcity of data and the weak correlation between geological and reservoir properties, often highlighted by the scattered point clouds of poro-perm diagrams. This is particularly true for complex reservoirs like carbonate reservoirs, whose heterogeneity results from multiple geological processes with different spatial organizations such as sedimentation and diagenesis (e.g. [Danquigny et al. 2019](#), [Rebelle 2014](#), [Ronchi et al. 2010](#)). In such contexts, establishing relevant geostatistical rules can prove to be a real challenge.

To cope with these issues, geologists and reservoir engineers rely on knowledge acquired and concepts developed over many years, particularly on outcrop and field analogues (e.g. [Howell et al. 2014](#), [Massonnat et al. 2017](#)). However, while most scientific disciplines attempt to derive conservation and constitutive laws from data and knowledge, and then establish physics-based models to reproduce and understand the observed phenomena, the distribution of reservoir properties is generally finally handled on a case-by-case basis using analogies, which is time consuming and unreliable. Many authors have certainly attempted to improve mathematical methods so that they produce more realistic models from a geological point of view (e.g. [Hu and Chugunova 2008](#)). This criterion of geological realism in models nevertheless appears in many publications (e.g. [Arnold et al. 2019](#), [Linde et al. 2015](#), [Rongier et al. 2014](#)), not as an end in itself or an ascertainable characteristic but as a means of reducing uncertainties and, undoubtedly, compensating for our inability to simulate geology until recently.

"Simulating geology" appears to be the crux of the problem. Although traditional geostatistical approaches honor measurements by interpolating between conditioning data, they lack a scientific basis for evaluating their predictive power. As per Popper's principles ([Popper 1963](#)), the only scientific way to validate models that are supposed to represent the result of deterministic geological interpretations is to simulate the controlling physical processes and to use data as a point of arrival rather than a point of departure, in order to narrow down the range of possibilities. This approach, which gives a physical meaning to property distribution, effectively reduces equifinality issues and enhances the predictive capacity of models ([Ebel and Loague 2006](#), [Willcox et al. 2021](#)).

Fortunately, the physical processes that shaped geological reservoirs are well understood. They are largely related to mechanics, with established conservative and constitutive laws that can be simulated numerically. For example, many current environmental studies rely on numerical simulation of sediment transport and deposition in a fluid environment, mechanical deformation and failure of rocks, or reactive transport of fluids in porous media. However, the spatial and temporal scale involved in simulating them at the reservoir or basin scale, and particularly at the geological time scale, is such that it has long been considered theoretically (upscaling) and numerically impossible.

To address this challenge, forward modelling methods for geology have developed significantly since the late 1990s. These approaches, whether they mimic processes with explicit time discretization (process-based methods) or produce spatial results of these processes according to an implicit chronology (rule-based methods), aim to preserve physical meaning and geological causality while achieving a necessary compromise between costly computational time and acceptable approximation. Some process-based methods are now sufficiently mature to accurately represent the geological heterogeneity of reservoirs (e.g. [Li et al. 2023](#), [Massonnat et al. 2021](#)). As well, while it remains computationally prohibitive to simulate all diagenetic processes by solving partial differential equations for reactive transport in porous media, it is possible to mimic their effects, such as the development of secondary or tertiary porosity ([Berra et al. 2016](#), [Hamon et al. 2021](#), [Lu et al. 2023](#), [Whitaker et al. 1999](#), [Fandel et al. 2022](#)). These methods are progressively replacing schematic handmade 1D/2D models, allowing to synthesize and compare our understanding of

geological processes in 3D, or even 4D by including temporal evolution. In doing so, they contribute to the maturation of geology as an experimental science (Burgess 2012).

Despite these advances, obstacles remain for the full application of forward modeling to simulate all the physical processes involved in the initiation and evolution of petrophysical properties over geological time, and thus build models of complex reservoirs for fluid dynamics simulations. To our knowledge, no single, comprehensive workflow exists to simulate the complete genesis of reservoir petrophysical properties. Process-based forward modeling typically stops at predicting geological characteristics like sedimentary facies, and almost never the properties needed to simulate the rest of the story. The distribution of petrophysical property is handled in a separate subsequent step, using geostatistics or simple mathematical relationships with a genetic characteristic such as grain size (e.g. Li et al. 2023). This breaks the causal link that these approaches aim to honor, in accordance with Popper's principles. This is especially problematic for carbonate reservoirs, where diagenetic and tectonic overprints with different spatial organizations can make it almost impossible to establish clear mathematical relationships between geological and petrophysical properties. It is typically in such cases that predictive modelling can prove particularly useful.

Predicting petrophysical properties therefore remains a challenge, particularly when these result from multiple interacting processes, such as sedimentation, compaction, diagenesis, fracturing, and karstification, which is the norm for carbonate reservoirs. These processes are linked to rock characteristics, such as mineralogy, that modelling should therefore initialize and vary. Some processes also depend on palaeoenvironmental conditions and fluid circulation, the effects of which can be particularly non-linear. Conversely, any evolution of mineralogy, porosity and permeability, among other factors affect the next steps, which can reinforce this non-linear behavior. To top it all off, these changes occur throughout geological history, from the deposition and early diagenesis phase (eogenesis), through burial (mesogenesis), to exhumation and late diagenesis (telogenesis).

In this paper, we propose a novel, global forward modeling workflow that jointly simulates the genesis of both geological and petrophysical properties of carbonate reservoirs. To maintain computational efficiency, this workflow focuses on the main processes affecting petrophysical properties evolution from deposition to the present day. These processes were identified in the literature and classified by diagenetic phase and their dependence on fluid circulation. We have redesigned our in-house forward modeling software suite to integrate this approach preserving the causal link between geological and petrophysical properties throughout the whole geological history. This enables the prediction of porosity and permeability as primary outputs. This work illustrates the methodology and, more importantly, opens the discussion on the conditions required to perform manageable, but truly predictive modeling of petrophysical properties in carbonate reservoirs. Our ultimate goal is to provide a fully forward solution that is computationally efficient enough to be applied at the reservoir scale.

Proposed solutions and workflow

As stratigraphic forward modeling is profoundly changing the way we do geology and fill geological models with depositional environments or facies (Borgomano et al. 2020), new rule- or process-based tools have been developed to simulate the diagenetic evolution of carbonate reservoir properties from deposition to present state (Lu et al. 2023, Whitaker et al. 1999, Fandel et al. 2022). This work is based on three existing tools respectively modeling sedimentology, fracturing and diagenesis related to fluid circulation in the porous medium, as karstification. The present work aims adapting these tools for a sequential application within a single, consistent workflow focusing on the main processes.

To preserve the chain of causality, the software was modified to integrate initial rock characteristics such as grain sizes and mineralogy utilized for calculating initial petrophysical properties rather than initiating them from the data. These properties are the starting point of the rock history. Given the high impact of early diagenesis, depositional processes, and synsedimentary diagenetic processes are simulated jointly rather

than sequentially. Finally, later diagenesis processes, such as karst formation, which require consideration of fluid flows in the reservoir, itself deformed and fractured during its structural history, are simulated. These key elements of the workflow that enables the prediction of petrophysical properties are presented below. For more details on how the various software programs involved work, readers can refer to other publications and patents (Massonnat, Gal, et al. 2025, Massonnat et al. 2023, Massonnat et al. 2022, Danquigny et al. patent application, 2023, Danquigny, Gal, et al. patent application, 2024, Danquigny, Leonforte, et al. patent application, 2024, Massonnat et al. patent application, 2024, Massonnat and Danquigny pending patent, 2022, Massonnat, Danquigny, et al. pending patent, 2022, Massonnat, Lemay, et al. pending patent, 2022, Danquigny, Massonnat, Ruiiu, and Dal Soglio pending patent, 2023, Danquigny, Massonnat, Ruiiu, Dal Soglio, et al. pending patent, 2023, Danquigny, Massonnat, Siffert, et al. pending patent, 2023, Massonnat, Danquigny, et al. pending patent, 2023, Massonnat, Leonforte, et al. pending patent, 2023).

Initiation of geological and petrophysical properties of deposits

Like any stratigraphic forward modelling tool, the applied software simulates the production, transport, and deposition on the seafloor of carbonate elements that will form the limestone and shape the reservoir. Specifically in this software, the basic elements, produced or introduced into the simulation, represent solid sediments whose characteristics are constrained more by reality and the objective of determining petrophysics through calculation than by the expected result in terms of lithology or thickness, which are calculated a posteriori. These elements thus have granulometric and mineralogical characteristics and are quantified in terms of mass. This quantification, as well as the definition of their characteristics, can be based on current environmental or ecological studies. More importantly, they make it possible to initiate and calculate the evolution of petrophysical properties.

The meshed model built by the algorithm is therefore made up of muddy and/or grainy sediments (following the Wentworth classification) and voids, which are divided into interparticular, moldic, and "vuggy" porosity. The first is initiated with the deposit, while the latter two can result from eogenesis. Immediately after the creation of each model cell, the initial porosity of the deposit is calculated based on the grain size distribution of the sediments and the percentage of grains, using an analytical formula calibrated using data from the literature (e.g. Enos and Sawatsky 1981). The initial permeability calculation is based on the analytical formula proposed by Lucia (2007), in which the Rock Fabric (RF) parameter is calculated from the grain size distribution of the sediments and the percentage of grains calculated by the software. A texture consistent with Dunham's classification (1962) and an initial compaction coefficient are also determined from the same parameters.

Eogenetic or syngenetic evolution of petrophysical properties

Numerous studies have demonstrated the predominant role played by early syn-sedimentary diagenesis in the development of petrophysical properties throughout the ensuing geological history (e.g. Leonide et al. 2014, Moore and Wade 2013a). Moreover, Lu et al. (2023) pointed out that these eogenetic processes also influence the evolution of sedimentary processes by impacting the thickness of deposits and thus topography, possibly before the overlying deposition of new sediments. For this reason, joint simulation of depositional processes and syn-sedimentary diagenetic processes is needed, especially in shallow marine carbonates.

All of the syngenetic processes identified in the literature have been synthesized into five main processes to be simulated with an analytical rule-based approach: mineralogical stabilization (i.e. replacement), dissolution, cementation, erosion, and compaction. While this is a simplification, these processes were selected as the primary drivers of petrophysical property evolution. The effects of these processes depend on environmental conditions, which are differentiated first according to hydrological zones (Jiao and Post 2019) and then subdivided into diagenetic zones (Moore and Wade 2013b), as illustrated in Fig. 1. At each time step, after quantifying the petrophysical properties of the upper meshes representing the deposits formed during that time step, the simulation determines first the hydrological zones, then the

diagenetic zone in which each mesh of the model is located, and calculates the effect of the corresponding processes accordingly.

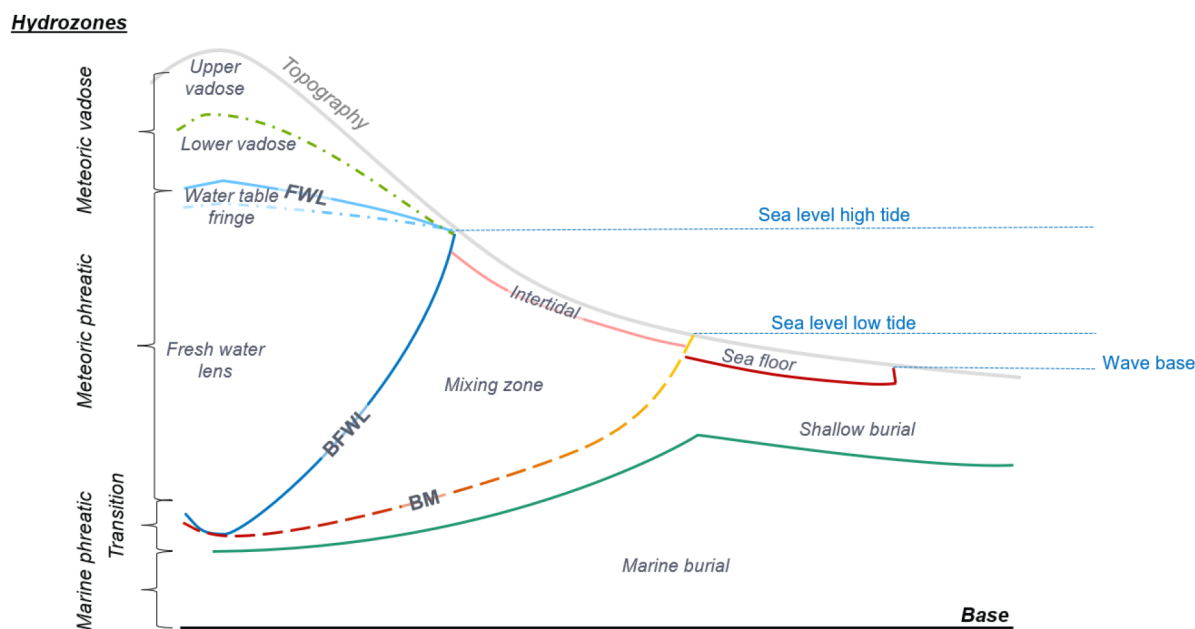


Figure 1—Schematic cross-section of the division into hydrological and diagenetic zones (FWL: Free Water Level, BFWL: Base of the Free Water Lens, BM: Base of the Mixing zone)

To determine the hydrological zones (Fig. 1), the emerged areas are first identified. Then, the boundaries of the freshwater lens are calculated by applying the Ghyben-Hetzberg equations (Jiao and Post 2019), which are functions of permeability and a rainfall recharge specified by the user. Finally, the other areas are delimited according to model parameters specified by the user such as sea level amplitude for the mixing zone. The diagenetic zones (Fig. 1) result from a sub-division by the modeler of the hydrological zones in order to specify the location of certain diagenetic processes. For example, the thickness of the upper vadose zone can be set at 30% of the total thickness of the vadose zone and that of the capillary fringe at 1 m. Dividing the vadose zone in two makes it possible to distinguish the positioning of dissolution and cementation processes within it. Similarly, dissolution processes in saturated zones occur preferentially in the capillary fringe.

For each diagenetic zone, the list of active processes was defined based on the literature (e.g. Moore and Wade 2013b). This list is presented in the companion paper of Massonnat, Barbier, et al. (2025). When several processes coexist in the same zone, their effects are calculated sequentially in an order that takes into account their respective kinetics and possible interactions in order to represent them as accurately as possible. For each process, in each zone, a simplified chemical kinetics, for example in the form of exponential decay, is applied according to the time step duration. The time constants for these kinetics are set by the user based on recommended ranges of values from the literature, which is, however, rather limited on such topics. Some values can also be defined relatively between zones. Other model parameters can also affect certain processes. For example, processes that depend on the intensity of fluid circulation are affected by the permeability or RF index.

Within our model, mineralogical stabilization and dissolution exclusively concern carbonates whose mineralogy distinguishes between aragonite, high-magnesium calcite (HMC), and low-magnesium calcite (LMC). Mineralogical stabilization simulates the dissolution of unstable minerals, such as aragonite, combined with the precipitation of more stable minerals, such as low-magnesium calcite, whose coupling

generates moldic porosity. Dissolution caused by fluid circulation generates a third type of porosity, known as "vuggy" porosity.

The other processes, cementation, erosion, and compaction, affect all lithologies and follow the same philosophy. Basic equations have been adapted from those in the literature according to variables that can be calculated by the software and a minimum of parameters requested from the user. These allow the user to adjust the sensitivity of the different processes to calibrate the result to the data. In the absence of equations in literature, analytical solutions have been calibrated on published data sets.

The related processes can profoundly modify porosity, from a scale smaller than that of the carbonate grain to that of the largest karst voids, with consequences on permeability that are equally difficult to assess or classify. Indeed, diagenesis gives rise to very different types of pore networks with very different dynamic behaviours (Leonide et al. 2014). To represent correctly the relationship between porosity and permeability, we thus need to distinguish between these different types of porosity, such as interparticular or vuggy porosity for example. This distinction allows for a better estimation of the resulting permeability. Indeed, each type of porosity has a different pore network topology, and therefore a potentially very different relationship between porosity and permeability (e.g. Catinat et al. 2023). Permeability thus is calculated using a different formula for each type of porosity, and the three are finally added together.

As a result, these developments enable to produce a forward deposit model populated with geological properties, as facies, but also porosity and permeability. Based on this result, it is possible to pursue the simulation of the following geological stages.

Modeling the tectonic evolution of the reservoir, as a prerequisite for simulating telogenesis

Among the diagenetic processes that affect carbonate reservoirs after the sedimentation and eogenesis phase, those resulting from fluid circulation, such as meteoric or hydrothermal fluids, are probably the most important in terms of their impact on petrophysical properties. Beyond the quantitative aspect, they can significantly alter the spatial organization of heterogeneity in ways that are difficult to predict. In order to best illustrate the approach adopted, the work presented here focuses on dissolution processes. These increase vuggy porosity, enlarge fractures and give rise to karstic features, such as conduit networks, thereby modifying not only porosity but, above all, the permeability field. Deformations and fracture networks resulting from tectonic processes primarily affect the flows that control this diagenesis. They must therefore be included in the modeling approach in the matrix and fractures of the reservoir, and even in preexisting karst conduits,

However, efficiently simulating the physical processes of deformation and fracturing remains a stumbling block for modelers. In the proposed modeling sequence, this part relies on rule-based techniques. In the one hand, the sedimentary model or the model from a previous phase is analytically deformed according to interpolation rules between the initial model and the current structural configuration (Danquigny et al. 2022). In the other hand, sets of fractures are generated with a Boolean modeling tool presented in more detail in the companion paper of Massonnat, Gal, et al. (2025). Thus, based on the deposit model, for each late diagenesis phase to be simulated, a model is generated that is deformed in the structural configuration at the corresponding geological time and explicitly includes discrete fracture networks.

Telogenesis related to fluids flows in the reservoir

Simulating diagenetic processes related to fluids circulation in different hydrological conditions, such as epigenic or hypogenic karst genesis, comes up against two difficulties: the need for flow simulation, which can be computational time consuming, and the difficulty of representing both the porous matrix, fractures and karst conduits together in the same model. To overcome them, a process-based approach that can mimic flows and reactive transport in both limestone matrix, fractures and karst has been developed.

In order to optimize the discretized representation of both the matrix, surface discontinuities such as fractures and inception horizons, and linear discontinuities such as karst conduits, respectively the cells,

certain faces and certain edges of a single polyhedral mesh are considered in a triple medium hybrid model. This mesh can result from merging a deposit model and a network of discrete fractures. The edges at the intersection of two polyhedral faces representing distinct fractures can also form the seeds of a network of karst conduits.

Each element considered in the mesh is thus associated with a void volume, parameterized by volume and porosity for volumetric meshes, area and aperture for faces, and length and radius for edges. For the matrix, the relationship between vuggy porosity and permeability is the same as that implemented for the simulation of eogenesis. For discrete elements, other formulas taken from the literature are applied (Király and Müller 1979, Snow 1965, 1969).

Then, a graph is constructed from the triple-medium hybrid model (Fig. 2). All active cells, faces, and edges of the hybrid model give rise to a node in the graph, connected to its closest neighbors. Regardless of the type of connected nodes, for each connection in the graph, an exchange surface and a transmissibility value according to a two-point formula (TPFA) are then calculated. The structure of the graph and its parameters thus make it possible to calculate a flow field using a finite volume solution method, which is simpler and less computationally expensive than other methods.

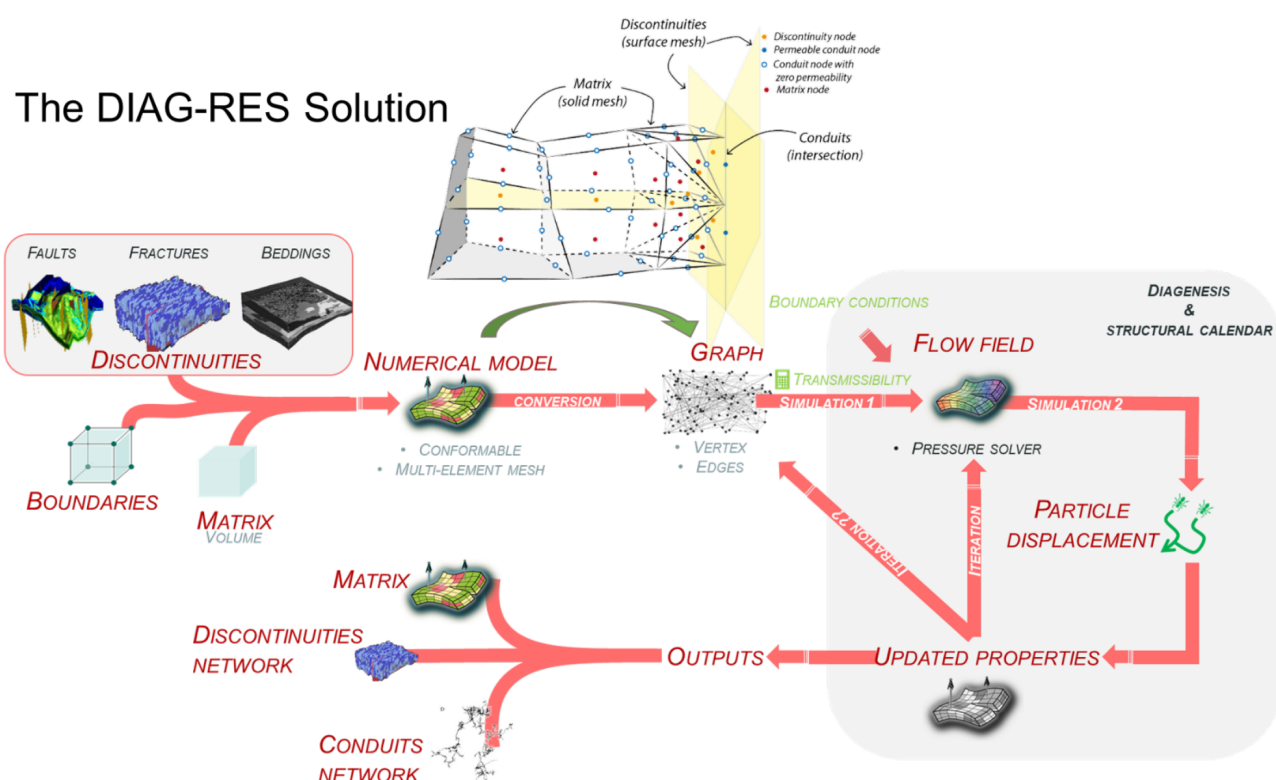


Figure 2—Workflow of the solution developed to simulate diagenetic processes related to fluid circulation in both matrix and discontinuities

Based on this calculated flow field, the reactive transport is simulated by a cellular automaton (Fig. 2), whose particles mimic the dissolution process, or any other evolution of the porous medium related to fluids flows, as mineralogical evolution or precipitation for example. Each particle injected into the model therefore represents a given volume of fluid and a given volumetric chemical reaction capacity, which depends on the assumed chemical characteristics of the input. The trajectory of the particles in the reservoir is determined probabilistically based on the transmissibility field and the pressure field determined previously. If dissolution is considered, the volume of rock dissolved by each particle is then distributed across the nodes along its trajectory based on the rock's sensitivity to dissolution. This sensitivity depends on facies

and/or texture, for example. Matrix nodes are affected directly based on their own sensitivity. On the other hand, for a fracture or conduit node, dissolution enlarges the element by trimming the adjacent matrix nodes, depending on their respective sensitivity, and therefore potentially in an anisotropic manner. To also take into account the evolution of the fluid's aggressiveness, decay rates based on the distance traveled or time elapsed have been implemented. This makes it possible to take into account both the nature of the dissolved rock and the evolution of the aggressiveness of the fluid on each trajectory. Finally, the void volume carried by each node is updated, resulting in a change in the petrophysical characteristics, and therefore in the transmissibility field and the flow field, between each time step.

Compared to previous calculation codes (e.g. [Jaquet et al. 2004](#)), this solution allows a better representation of the flow field and therefore of the paths traveled by the particles, regardless of the different media, which are also represented more finely thanks to two- and one-dimensional discrete elements and polyhedral meshing. This approach therefore corrects the shortcomings of previous tools and should produce karst networks with a topology comparable to real networks. Finally, the evolution of the volume of voids resulting from dissolution processes ensures that the mass balance is preserved in relation to time discretization. This paves the way for quantitative analyses of the conditions for the development of porosity and karst.

Results

Two models were selected to illustrate the main stages of the proposed workflow and validate the concept. The first model highlights the petrophysical heterogeneity of carbonate reservoirs immediately after their deposition, resulting from the combination of different sedimentary facies and successive environmental conditions that controlled early diagenesis. The second model highlights how karstification processes also increase the heterogeneity of carbonate reservoirs.

Deposit model of a Tethyan carbonate platform

The geological context of the synthetic model presented here is largely inspired by that of the Barremian-Aptian Urgonian platform in southeastern France. The model reproduces a carbonate platform deposited between 128 and 126 million years ago (MyA), covering an area of 50×70 km. It comprises 80 layers of 25 ky and 200×280 cells per layer. Ten carbonate elements are produced by the model: green algae, gastropods, rudists, corals, echinoderms, sponges, peloids, ooids, neritic mud, and pelagic mud. For each element, the production rate and the transport of produced sediments depend mainly on the grain size, bathymetry and hydrodynamic energy. The two latter are calculated by the sedimentary forward simulation. The relationships between these parameters are taken from the database linked to the software, which is based on literature and TotalEnergies' expertise in carbonate sedimentology (e.g. [Michel et al. 2023](#)). While the sea level curve comes from [Haq \(2014\)](#), the main other assumptions entered into the model are the initial topographic map and subsidence values, ranging from -40 m to 400 m/Ma and distributed according to space and time. The latter result from the interpretation of lithological logs acquired in the field. The result is therefore very close to a real-life case.

The [Fig. 3](#) and [Fig. 4](#) illustrate some of the calculated outputs of the forward simulation, respectively geological and petrophysical properties. The distribution of sedimentary facies is, quite logically, more heterogeneous than the various geological characteristics taken separately and which the facies aim to synthesize. Similarly, it is consistent with the distribution of initial petrophysical properties. However, despite the synthetic nature of facies and the fine heterogeneity that sedimentary simulation is able to produce, the overall organization of petrophysical properties is particularly different at the end of the sedimentary and eogenetic sequence.

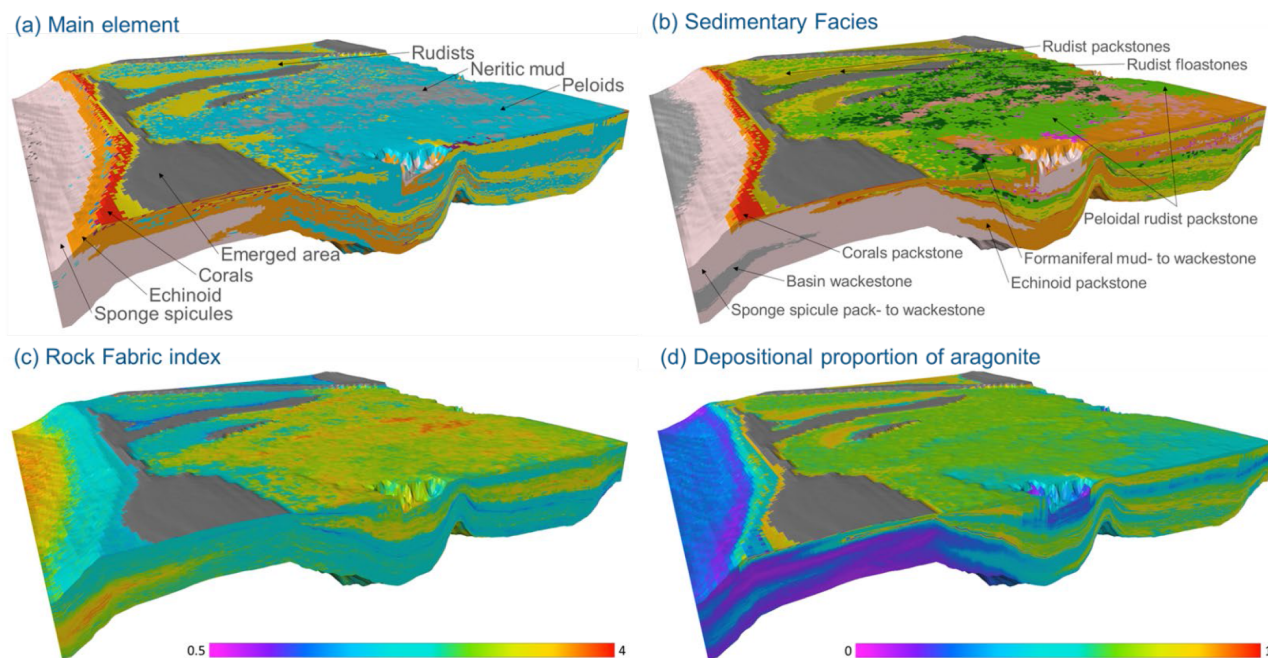


Figure 3—Examples of geological properties predicted by the process-based sedimentary simulation: (a) basic element mainly deposited in each cell of the model; (b) Sedimentary facies calculated from the descriptive facies table entered by the user and the characteristics of the elements deposited by the simulation; (c) Rock Fabric index (Lucia 2007) calculated from the distribution of grain sizes; (d) Proportion of aragonite at depositional time.

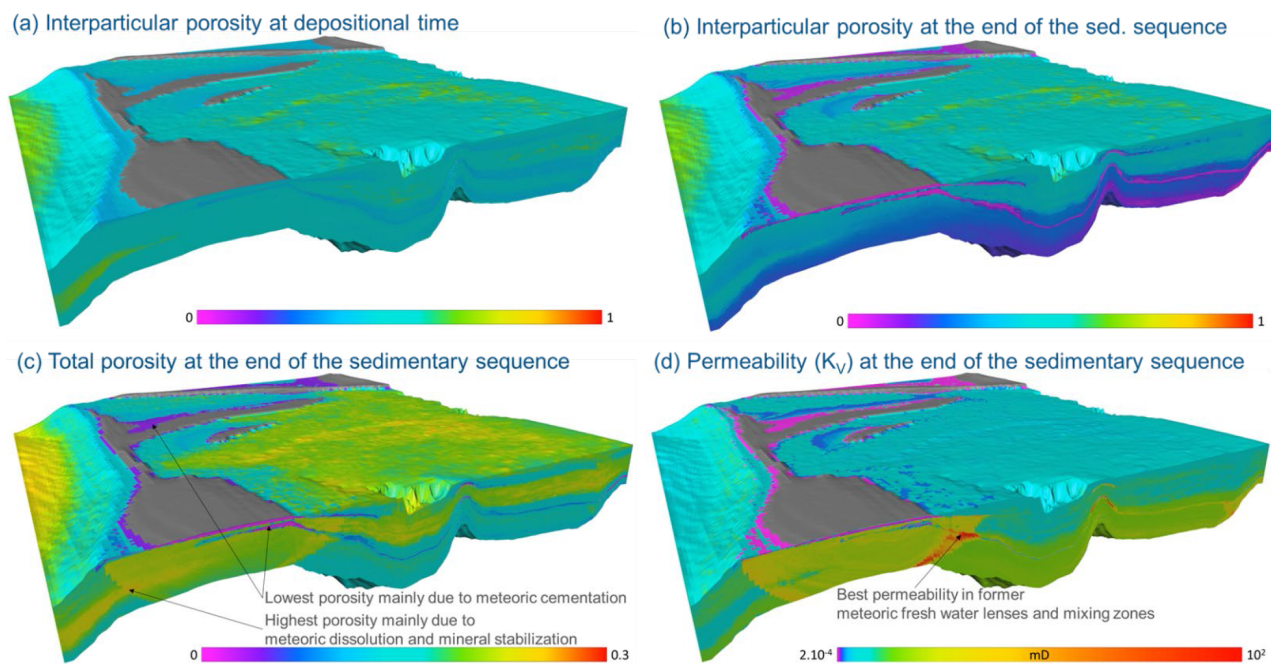


Figure 4—Examples of petrophysical properties predicted by the coupled process-based sedimentary and eogenetic simulation: (a) interparticular porosity at depositional time; (b) interparticular porosity at the end of the sedimentary and eogenetic sequence; (c) Total porosity summing interparticular, modic and vuggy porosities at the end of the sedimentary and eogenetic sequence; (d) Vertical permeability at the end of the sedimentary and eogenetic sequence.

These results illustrate the difficulties typically encountered in inferring petrophysical properties solely from sedimentary facies. They also confirm the value of jointly simulating sedimentary and diagenetic processes to predict these properties.

Karst model

A synthetic model was constructed to illustrate the impact of simulated dissolution processes only. It consists of four folded layers delimited by three inception horizons, with an extent of $17 \times 9 \times 5$ km. Three subvertical faults complete the structural model. Each horizon initially has almost homogeneous Gaussian distributions of porosity and permeability to best highlight the effect of the simulation. The simulation reproduces a short sequence (50 ky) of epigenetic karstification. Uniform recharge feeds the reservoir, infiltrates, and flows toward its right boundary.

The Fig. 5 illustrates the structural model, some trajectories of particles injected into the model, and some resulting modified properties, namely the porosity and permeability of the matrix and the aperture of discontinuities at the end of the simulation. It reveals that, even though initially the porous matrix in each horizon is almost homogenous and the aperture of discontinuities and the recharge are uniform (1 mm), the dissolution processes induced by groundwater flows develop petrophysical heterogeneity through evolving preferential flow paths in both the porous matrix and the discontinuities. In this example, although there is no karst conduit at the start of the simulation, a vertical conduit appears in the results.

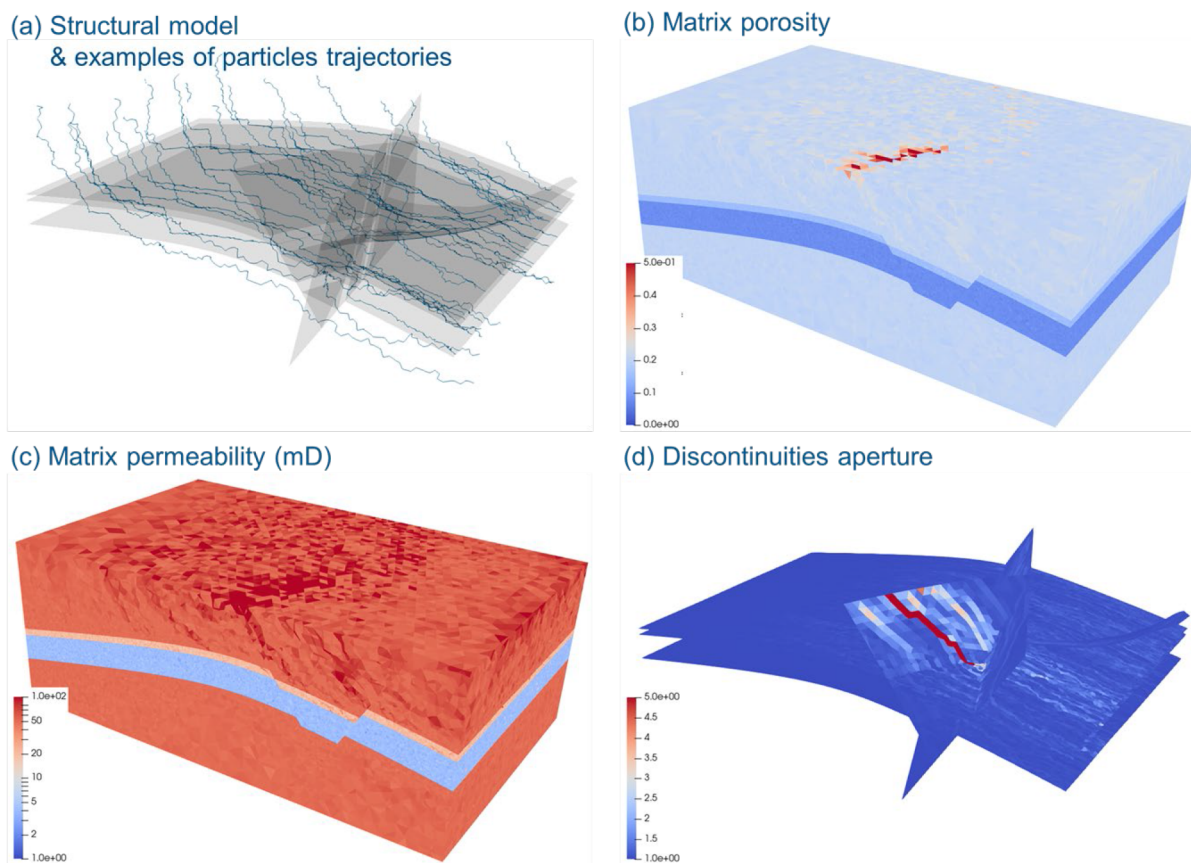


Figure 5—Results of the simulation of dissolution caused by fluid circulation in the porous medium on a synthetic model: (a) Structural model and examples of particles trajectories; (b) Matrix porosity at the end of the simulation (the four layers were initially homogeneous); (c) Matrix permeability at the end of the simulation (the four layers were initially homogeneous); (d) Aperture of the discontinuities at the end of the simulation (inception horizons and fractures initially have uniform aperture).

This model highlights the considerable impact that successive paleo-flow fields can have on current petrophysical heterogeneity and, consequently, the importance of taking them into account in the distribution of petrophysical properties in the reservoir model. Intuitively, this seems very difficult without time-dependent modeling.

Obviously, the resulting heterogeneity is greater and more realistic when the initial model, resulting from the succession of sedimentary-eogenetic simulations and fracturing simulations, is itself more realistic and heterogeneous.

Discussion

The models presented here remind us that, despite its complexity, the geological history of a reservoir is the key factor in predicting the distribution of its properties in a model. Successfully simulating the related processes according to computational capabilities poses a problem of upscaling, whose complexity is mainly due to the magnitude of the time scale factor. The development of predictive simulation tools for geology, particularly stratigraphy, sedimentology, and diagenesis, has been attempting to address this upscaling for about 25 years. The work presented in this article demonstrates that these tools and their development methodology are reaching an encouraging stage in their development. We have thus produced promising tools for simulating sedimentology, eogenesis, fracturing, and telogenesis, some of which are presented in this document. Coupling them in a single workflow offers the prospect of being able to build geological reservoir models filled with petrophysical properties entirely from process-based simulations.

In addition to the tools developed, this work also establishes a methodological approach for developing geological process-based forward modelling methods, which consider both the main processes to be reproduced in order to obtain the desired result and the data generally available. Indeed, this methodology incorporates as much as possible the results of the literature and observations gathered in the field, particularly in the form of equations and quantified parameters. It seeks also to optimize the number and nature of the parameters requested from the user.

For each geological process to be simulated, this methodology is as follows: a bibliographic review is first conducted in order to identify and characterize the main biophysical and chemical processes involved; then, for each process considered, a mathematical model is selected from among those identified, based on its assumptions, its degree of approximation, and its scalability to geological time scales; if no suitable model is available, an analytical solution is calibrated to the data sets to reproduce the observed trends; next, coupling rules are established for synchronous processes. The choice and development of the selected solutions also incorporates two constraints: input parameters with a physical meaning for the user, whose order of magnitude can be quantified from the literature or in the field, and numerical efficiency. This framework thus sparked the adaptation of characterization methods to produce, in addition to measurements to be compared with the model's output variables, input measurements that can be calibrated with the simulations (e.g. [Michel et al. 2023](#), [Barbier et al. 2025](#)). To meet the second constraint, most of the proposed solutions use particle tracking methods to mimic the effects of fluid circulation ([Delay et al. 2005](#)). The accuracy of these methods is adapted to the weight of these flows in the mimicked process.

Despite its simplifications, the package provides therefore a development framework for predicting the values and distribution of geological and petrophysical properties in a model of a current reservoir and the possibility of validating them by comparison with observations and measurements. To achieve this, we had to make choices between the accuracy of the simulated physics and computation time. But, this may evolve with the development of computing capacities. We have also taken advantage of the literature and existing databases, which remain nevertheless rather limited on the quantification of some processes. Some analytical relationships still need to be refined and calibrated on case studies to be increasingly predictive in terms of reservoir properties. Because the software suite is both fully forward and not too computationally time consuming to apply at the reservoir scale, it offers an interesting framework to challenge not only field measurements and the concepts proposed by geologists but also its own algorithm.

Conclusion

The tools and the workflow presented in this paper succeed in taking into account the main diagenetic processes affecting carbonate reservoirs to calculate petrophysical properties. Initial simulations have produced realistic and promising results. Because it preserves geological causality and not simply interpolates between conditioning data, this new approach opens the door to actually predict the present days petrophysical properties of carbonate reservoirs. They contribute to move geological process-based forward modeling both from modeling geological properties to simulating petrophysical properties, and from multiple tools with different constraints to a single integrating software capable of simulating all the processes at the origin of petrophysical heterogeneity. This therefore disrupts conventional workflows and allows the data to be used to falsify modeling assumptions, reduce large initial uncertainty and calibrate the prediction.

Furthermore, while purely mathematical approaches provide concrete solutions to engineering problems, they also create the illusion of generating data. In this respect, they can hinder understanding, which they claim to circumvent, whereas physics-based models open the door to understanding. In particular, solutions such as those proposed provide a framework that facilitates the coupling and alignment of diverse data.

Consequently, everything is in place to begin simulating and validating the heterogeneity of reservoir models, just as fluid dynamics are finally simulated and validated in these models. This work thus contributes to the evolution of geology from a natural science to an experimental science, in Popper's sense, or in other words, toward quantitative geology. The calibration of certain processes on one model, such as the Vacluse reservoir in southeastern France, can thus feed into the next ones, such as those of other karst aquifers in southeastern France or cretaceous reservoirs in the Middle East, or call into question the interpretations of measurements. This approach therefore also becomes a means of improving data interpretation and fundamental understanding of the genesis of the objects of study through model calibration. In turn, it will finally be possible to improve and continue the development of the proposed solutions. The developed framework allows such an exciting work.

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